ARTIFICIAL INTELLIGENCE AND THE LESSONS OF HISTORY

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ABSTRACT

The history of technological advancement is often invoked as a reason for optimism about the future of artificial intelligence. AI supporters infer from history that regulating AI would be counterproductive, because new technologies are typically beneficial to society overall. Those who think AI is unlikely to have much real-world impact also invoke technological history, pointing to prior inventions that failed to live up to their initial hype. Yet the lessons of technological history are not as clear as many believe. This Essay examines several historical episodes where elites made grave errors involving new technologies and threats, sometimes due to excessive optimism and sometimes due to excessive skepticism. History reveals that novel technologies can exceed expectations, have harmful effects on society, cause serious environmental harms, and even risk global catastrophe. Policymakers and regulators should find little basis for complacency about AI in technological history.

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INTRODUCTION

Technological history is often invoked as a reason for optimism about the future of artificial intelligence. People were afraid of the industrial revolution and the printing press, and those turned out to be remarkably beneficial to human civilization.[1] AI optimists infer from this history that we should not worry so much about AI's effects and that everything is likely to work out as it often has before.[2]

Technological history is also invoked by those who think that AI is unlikely to transform society much at all.[3] History is replete with hype cycles surrounding exciting new technologies that turned out to have only minor impacts on the world.[4] Indeed, relatively few promising early-stage technologies end up becoming useful, let alone transformative. AI might be just another technology that turned out to be less significant, less capable, and accordingly less dangerous, than initially believed.

Yet the lessons of technological history are not always so comforting. There are many examples of decisionmakers being wrong about new technologies and potential threats, sometimes due to excessive skepticism and sometimes excessive optimism. Throughout history, skeptics have often

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[1] *See* Joel Mokyr et al., *The History of Technological Anxiety and the Future of Economic Growth: Is This Time Different?*, 29 J. ECON. PERSP. 31, 33–38 (2015); Frank Furedi, *Moral Panic and Reading: Early Elite Anxieties About the Media Effect*, 10 CULTURAL SOC. 523, 527–28 (2016).

[2] *See, e.g.*, Mark Andreessen, *The Techno-Optimist Manifesto*, ANDREESEN HOROWITZ (Oct. 16, 2023), https://a16z.com/the-techno-optimist-manifesto/; Brett A. Hurt, *A Broader Grasp of Technology's History is Key to the Bright Future of AI*, MEDIUM (Oct. 30, 2024), https://databrett.medium.com/a-broader-grasp-of-technologys-history-is-key-to-the-bright-future-of-ai-1c78bc0c5d6e; Dunan Weldon, *History Suggests the AI Backlash Will Fail*, TRANSFORMER (Nov. 6, 2025), https://www.transformernews.ai/p/history-suggests-the-ai-backlash-luddities-printing-press-jobs-displacement.

[3] *See, e.g.*, Arvind Narayanan & Sayash Kapoor, *AI as Normal Technology: An Alternative to the Vision of AI as a Potential Superintelligence* 5–10, KNIGHT FIRST AMEND. INST. (Apr. 15, 2025), https://knightcolumbia.org/content/ai-as-normal-technology; Debra Howcroft & Phil Taylor, *Automation and the Future of Work: A Social Shaping of Technology Approach*, 38 NEW TECH. WORK & EMP. 351, 352–53 (2023); Chris Vallance, *Meta Scientist Yann LeCun Says AI Won't Destroy Jobs Forever*, BBC NEWS (June 15, 2023), https://www.bbc.com/news/technology-65886125.

[4] *See, e.g.*, TOI World Desk, *10 Promising Inventions that Failed to Change the World*, TIMES OF INDIA (Sep. 4, 2024), https://timesofindia.indiatimes.com/world/10-promising-inventions-that-failed-to-change-the-world/articleshow/113054403.cms; M6 Labs, *From Cultural Phenomenon to Poverty - How NFTs Failed*, CRYPTO PRAGMATIST (Oct. 30, 2024), https://cryptopragmatist.com/p/from-cultural-phenomenon-to-poverty-how-nfts-failed.

underestimated the likelihood of novel innovations and their potential ramifications for humanity. Others have been overly optimistic about the social effects of new technologies or the strategic benefits of racing to build dangerous new weapons. Virtually every transformative technology, from the printing press to the automobile to the smartphone, was underestimated or misunderstood in its early stages.[5] And for most major disasters, from pandemics to hurricanes, there have been a host of early skeptics who prove tragically wrong.[6] This Essay examines a set of historical episodes where technological skepticism or optimism was mistaken, not to prove that AI skepticism or AI optimism is necessarily mistaken, but to demonstrate that history is not definitive in any direction. Indeed, history demonstrates that that new technologies can sometimes exceed expectations, have harmful impacts on society, cause serious environmental damage, and even risk global catastrophe.

The history of new technologies and their potential risks bears directly on debates surrounding the law's approach to artificial intelligence. If concerns about the impact of new technologies are always overblown, regulating new technologies with substantial upsides like AI would likely be counterproductive. If AI is unlikely to dramatically improve on its current capabilities or radically change its structure or design, then concerns about catastrophic risk and regulations to address such risks are likely unnecessary. By contrast, if the history of technology is mixed as to the development and the potential risks of new technologies, expert concerns about AI risks should be taken more seriously.[7] Indeed, history suggests that new technologies do not inevitably benefit society, and that taking steps to

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[5] James Clive-Matthews, *A Brief History of Tech Skepticism*, STRATEGY + BUS. NEWS. (May 5, 2023), https://www.strategy-business.com/article/A-brief-history-of-tech-skepticism; ELIZABETH L. EISENSTEIN, THE PRINTING PRESS AS AN AGENT OF CHANGE: COMMUNICATIONS AND CULTURAL TRANSFORMATIONS IN EARLY-MODERN EUROPE (1997).

[6] Evan Bush & Randi Richardson, *Storm forecasts have never been more accurate. Meteorologists say they've never faced so much pushback*, NBC NEWS (Oct. 16, 2024), https://www.nbcnews.com/science/science-news/meteorologists-accurate-storm-forecasts-skepticism-pushback-rcna175403; Ryan Benk et al., *Skepticism Of Science In A Pandemic Isn't New. It Helped Fuel The AIDS Crisis*, NPR (May 23, 2021), https://www.npr.org/2021/05/23/996272737/skepticism-of-science-in-a-pandemic-isnt-new-it-helped-fuel-the-aids-crisis.

[7] *See* Katja Grace et al., *Thousands of AI Authors on the Future of AI*, 84 J. ARTIFICIAL INTELLIGENCE RES. 9:1, 9:10 (2025) (reporting an extensive survey of experts with the median expert estimating a ten percent chance that AI will lead to humanity's extinction or its equivalent).

systemically regulate AI, mitigate its potential harms, and address its potentially catastrophic future outcomes may be justified. This Essay and the historical episodes it explores demonstrate that the future of new technologies is unpredictable, and the risks of such technologies are potentially serious. Policymakers and regulators should find little basis for complacency about AI in the relevant history.

The Essay proceeds as follows. Part I surveys historical episodes where excessive skepticism or excessive optimism about new technologies led experts, pundits, or policymakers to make serious errors in their predictions about the future. Part II explores a recent example of mistaken and prolonged skepticism about a major global risk, and examines the similarities to modern debates about AI risks. Part III gives a detailed account of the Cold War arms race between the United States and the Soviet Union, examines the strategic impacts and downside risks of building thermonuclear weapons, and draws lessons for today's policymakers about the potential implications and harms of superpower arms races.

I. HISTORICAL PREDICTIONS CONCERNING TECHNOLOGICAL DEVELOPMENT

New technologies are often overhyped by marketers and early adopters, and many initially promising technologies wind up having little impact on the world.[8] In other cases, experts and pundits are excessively skeptical of new technologies that come to transform society on a global scale. Optimistic projections of new technologies' impacts on society also frequently go wrong. Indeed, historical predictions about the development and impact of new technologies are highly error prone in all directions. This section gives two examples of historical instances where excessive skepticism or excessive optimism about new technologies led to erroneous predictions about the future.

A. Nuclear Fission

The discovery of controlled nuclear fission in 1938 came as a shock.[9] It

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[8] *See, e.g.*, Kornelia Konrad et al., *Strategic Responses to Fuel Hype and Disappointment*, 79 TECH. FORECASTING & SOC. CHANGE 1084 (2012).

[9] Lawrence Badash et al., *Nuclear Fission: Reaction to the Discovery in 1939*, 130 PROC. AM. PHIL. SOC. 196, 199 (1986).

was not an expected next-step in nuclear research; it was a breakthrough that many believed would never happen.[10] In the decades prior to its discovery, controlled nuclear fission was considered by many experts to be fodder for science fiction, rank speculation, or pure "moonshine."[11] Some experts were annoyed by speculative newspaper stories in the early 1930s hyping the potential for a new and wonderful nuclear age based on a few early experiments.[12] People had been making conjectures about atomic or radiation-based energy, and about the weapons that such energies might enable, since the turn of the century, and these speculations had never panned out.[13]

Many prominent physicists and other experts cast doubt on the prospect of controlled nuclear fission ever being achieved. Albert Einstein himself confidently predicted that nuclear energy would never be harnessed by man. In a 1934 talk at the Carnegie Institute of Technology, he compared unlocking nuclear power to "shooting birds in the dark in a country where there are only a few birds," deeming it impractical for energy generation.[14] Looking into the distant future, he declared that "[t]here is not the slightest indication that [nuclear] energy will ever be obtainable."[15] Niels Bohr, one of the foremost physicists of the 20th century, expressed his skepticism of nuclear energy in 1936, and opined that recent research into nuclear reactions suggested that harnessing such energy appeared to be a remote prospect, at

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[10] *Id.*

[11] *E.g.*, *id.* at 197 (quoting eminent physicist Ernest Rutherford in 1933). Rutherford's remarks at the 1933 meeting of the British Association for the Advancement of Science adamantly asserted that the idea of harnessing nuclear fission energy was science fiction. *Id.* at 199. The *New York Herald Tribune* quoted several other prominent nuclear scientists agreeing with Rutherford in a story entitled "Atom-Powered World Absurd, Scientists Told." *See id.* at 203. Rutherford had previously cast doubt on the possiblity of mankind developing powerful atomic bombs. He told a fellow chemist that "You need not be alarmed about any possibilities of atomic disintegration [leading to large scale explosions]; if it had been feasible it should have happened long ago on this ancient planet. I sleep quite soundly at nights." *Id.* at 200. Echoing these sentiments, Robert A. Millikan of Caltech rejected calls for a moratorium on nuclear research, believing that mankind was incapable of building nuclear technologies that could potentially threaten the world. He was convinced "that the creator has put some foolproof elements into his handiwork and that man is powerless to do any titanic physical damage." *Id.* at 202.

[12] Badash, et al., *supra* note 9, at 197.

[13] *Id.* at 198–99.

[14] *Atom Energy Hope is Spiked by Einstein*, PITTSBURGH POST-GAZETTE, Dec. 29, 1934. at 1.

[15] *The Einstein Letter That Started It All; A message to President Roosevelt 25 Years ago launched the atom bomb and the Atomic Age*, NEW YORK TIMES, Aug, 2, 1964, at 169.

best.[16] Robert Oppenheimer was so skeptical about nuclear fission that when the physicist Luis Alvarez told him in early 1939 that German scientists Otto Hahn and Fritz Strassman had discovered fission in their lab, Oppenheimer ran to his blackboard and "proved" that fission was impossible.[17] Oppenheimer didn't change his mind until the next day, when Alvarez produced nuclear fission in his laboratory and showed it to Oppenheimer, who immediately began to speculate about how the process could be used to make reactors and bombs.[18] Numerous other prominent scientists of the day were similarly skeptical that controlled fission would ever be discovered, in the years just before it was discovered.[19]

Albert Einstein, Neils Bohr, and their fellow skeptics obviously did not fail to anticipate nuclear fission due to a lack of intelligence. They simply failed to recognize that a variety of smart people all around the world trying different approaches and building on each other's successes were capable of unforeseeable breakthroughs. Even Hahn and Strassman did not expect to produce controlled fission before doing so during an experiment designed to study non-fissile nuclear activity.[20] We can observe something similar in contemporary discussions of AI capabilities. Much of the skeptical discourse around AI's potential risks implicitly assumes that there will be no further breakthroughs in AI capabilities on the level of the "transformer" that enabled modern Large Language Models.[21] But tomorrow's AIs are likely to be fundamentally different from today's. The exact nature of future breakthroughs may be difficult to foresee from the vantage point of the present, just as nuclear fission was in 1936. But breakthroughs may occur, and may transform the world, nonetheless.

Predicting the future of technological development is difficult even in the short term. The fission skeptics were well-versed in the limitations of

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[16] Badash, et al., *supra* note 9, at 203; Neils Bohr, *Neutron Capture and Nuclear Constitution*, 137 NATURE 348 (1936).

[17] Badash, et al., *supra* note 9, at 211; KAI BIRD & MARTIN J. SHERWIN, J. ROBERT OPPENHEIMER, AMERICAN PROMETHEUS: THE TRIUMPH AND TRAGEDY OF J. ROBERT OPPENHEIMER 166 (2007).

[18] Badash, et al., *supra* note 9, at 211; BIRD & SHERWIN, *supra* note 1717, at 166.

[19] *See* Badash et al., *supra* note 9, at 197–204 (stating that these scientists included Arno Brasch, A.S. Eddington, H.C. Dickenson, I.I. Rabi, Victor La Mer, and others).

[20] Alan Chodos, *December 1938: Discovery of Nuclear Fission*, APS123 (Dec. 3, 2007), https://www.aps.org/apsnews/2007/12/december-1938-discovery-nuclear-fission.

[21] *See, e.g.*, Tapio Pitkäranta, *Forecasting Surprises in Machine-Learning-Driven Interaction Systems: Lessons from the Transformer Breakthrough*, *in* COMPUTER-HUMAN INTERACTION RESEARCH AND APPLICATIONS, CHIRA 2025 165 (Josef F. Krems, Hugo Plácido da Silva & Pietro Cipresso eds., 2026).

nuclear science and the virtual impossibility of splitting atoms under existing theoretical paradigms. They had an encyclopedic knowledge of what existed, and little grasp of what did not yet exist. It seemed obvious only in retrospect. As Bohr exclaimed when told about the discovery of nuclear fission, "Oh, what fools we have been! We ought to have seen that before."[22]

B. The Internet

In 1989, British scientist Tim Berners-Lee developed HyperText Markup Language (HTML), which became the basis for the "World Wide Web" and the modern internet.[23] The number of internet-connected computers grew from two thousand in 1985 to more than two million in 1993.[24] By 1995, Yahoo!, Internet Explorer, AOL, eBay, Craigslist, and Amazon were online.[25] Google search arrived in 1998, at a time when about a third of all Americans were already connected to the internet.[26]

Yet skepticism of the internet and its economic and cultural significance persisted throughout this early era, ceasing only when the internet had become such a major part of daily life that its impact was undeniable. Famously, Nobel-Prize-winning economist Paul Krugman predicted in 1998 that the growth of the Internet would soon slow drastically, and that "the Internet's impact on the economy would be no greater than the fax

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[22] Badash et al., *supra* note 9, at 208.
[23] Gayla Koerting, *Development of HTML*, EBSCO (2023), https://www.ebsco.com/research-starters/history/development-html.
[24] Kim Ann Zimmermann & Jesse Emspak, *Internet history timeline: ARPANET to the World Wide Web*, LIVE SCI. (Apr. 8, 2022), https://www.livescience.com/20727-internet-history.html.
[25] Jon Skillngs, *In '95, These People Defined Tech: Gates, Bezos, Mitnick and More*, CNET (May 27, 2020), https://www.cnet.com/tech/computing/in-95-these-people-defined-tech-gates-gosling-bezos-mitnick-and-more/; Alyssa Newcomb, *Yahoo, AOL and a Look Back at Other 90s Internet Stars*, NBC NEWS (Apr. 4, 2017), https://www.nbcnews.com/tech/tech-news/yahoo-aol-gone-here-s-look-back-internet-graveyard-n742551; *How Internet Explorer Took Over the Early Online World, and Where it is Today*, MICROSOFT (Feb. 13, 2024), https://www.microsoft.com/en-us/edge/learning-center/how-internet-explorer-once-took-over-the-web?form=MA13I2.
[26] Max Roser, *The Internet Has Just Begun*, OUR WORLD IN DATA (Oct. 3, 2018), https://ourworldindata.org/internet-history-just-begun.

machine's."[27] This turned out not to be accurate.[28] But Krugman was hardly alone; 1990s-era internet skepticism could be found in corporate suites, existing tech companies, and across print media. A 1997 New York Times article quoted a software company CEO who noted the relatively small amount of money transacted on the internet and doubted that it would ever increase much.[29] The internet would never replace physical stores or print newspapers, he claimed.[30] Countless thinkpieces in Newsweek, Wired, and other magazines predicted the imminent failure of the internet.[31] The inventor of Ethernet technology predicted in 1995 that the internet would collapse by the next year.[32]

For the most part, predictions of the internet's collapse or the leveling off of its growth were not inherently unreasonable. Rather, whether a promising technology will become transformative, the future course of a transformative technology in its early stages, and the point at which its growth will slow, are difficult to predict. Some technologies fail to live up to even a fraction of their early promise. Others exceed all but the wildest early expectations.

A different set of internet predictions focused on its future social impacts. In the early years of the internet, the prevailing optimist view was that the Internet would promote democracy and undermine authoritarian governments.[33] Speaking in 2000, President Bill Clinton expressed this

[27] David Emery, *Did Paul Krugman Say the Internet's Effect on the World Economy Would Be 'No Greater Than the Fax Machine's'?*, SNOPES (Jun, 7, 2018), https://www.snopes.com/fact-check/paul-krugman-internets-effect-economy/; Paul Krugman, *Why Most Economists' Predictions are Wrong*, RED HERRING ONLINE, https://web.archive.org/web/19980610100009/http:/www.redherring.com/mag/issue55/economics.html.

[28] *See, e.g.*, STEVEN D. LEVITT & STEPHEN J. DUBNER, THINK LIKE A FREAK: SECRETS OF THE ROGUE ECONOMIST (2014).

[29] Sharon Reier, *Skeptics Cite Overload Of Useless Information : Internet Arrives At a Crossroads*, N.Y. TIMES (Mar. 14, 1997), https://www.nytimes.com/1997/03/14/news/skeptics-cite-overload-of-useless-information-internet-arrives-at-a.html.

[30] *Id.*

[31] Amelia Tait, *25 Years On, Here are the Worst Ever Predictions About the Internet*, NEW THINKING (Apr. 4, 2022), https://www.newstatesman.com/science-tech/2016/08/25-years-here-are-worst-ever-predictions-about-internet.

[32] *Id.*

[33] *See, e.g.*, Joseph Nye, *Protecting Democracy in an Era of Cyber Information War*, HOOVER INST.: GOVERNANCE IN AN EMERGING NEW WORLD (Nov. 18, 2018), https://www.hoover.org/research/protecting-democracy-era-cyber-information-war; Farid Shirazi, *The Contribution of ITC to Freedom and Technology: An Empiricial Analysis of Archival Data on the Middle East*, 35 ELEC. J. INFO. SYS. DEV. COUNTRIES 1 (2008).

popular view, predicting that the internet would help to liberalize China and stating that China's early efforts to crack down on internet expression were "sort of like trying to nail Jell-O to the wall."[34] Numerous pundits and experts shared this view, contending, for instance, that "[a]s more and more people have access to the Internet, it will be practically impossible for governments to ban something."[35] Others suggested that the internet would promote equality and egalitarianism, decentralizing political power.[36]

These predictions about the internet's anti-authoritarian and pro-democracy effects have largely proven incorrect. In authoritarian countries, the internet has facilitated extensive monitoring of citizens' activities and beliefs, and censorship of their expression.[37] In particular, the Russian and Chinese governments have skillfully leveraged internet monitoring and other data collection systems to reinforce their rule and deter dissent.[38] Because a substantial amount of modern political deliberation and organization occurs online, it is often more visible to authoritarian regimes and more susceptible to censorship and disruption.[39] Even in more democratic countries, the internet has allowed governments to conduct extensive and low-cost surveillance of their populations through digital means, potentially chilling expression or facilitating government abuses.[40] Rather than democracies

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34 William J. Clinton, *Remarks at the Paul H. Nitze School of Advanced International Studies,* THE AMERICAN PRESIDENCY PROJECT (March 8, 2000), https://www.presidency.ucsb.edu/documents/remarks-the-paul-h-nitze-school-advanced-international-studies.

35 Peter H. Lewis, *Ideas & Trends; On the Internet, Dissidents' Shots Heard 'Round the World*, N.Y. TIMES (June 5, 1994), https://www.nytimes.com/1994/06/05/weekinreview/ideas-trends-on-the-internet-dissidents-shots-heard-round-the-world.html.

36 John Perry Barlow, *A Declaration of the Independence of Cyberspace*, EFF (Feb. 8, 1996), https://www.eff.org/cyberspace-independence.

37 *See* FREDERICO MANTISELLI, DIGITAL AUTHORITARIANISM: HOW DIGITAL TECHNOLOGIES CAN EMPOWER AUTHORITARIANISM AND WEAKEN DEMOCRACY (Geneva Cent. Sec. Pol'y Feb. 16, 2023), https://www.gcsp.ch/sites/default/files/2024-12/in-focus6-digital-authoritarianism.pdf.

38 *See, e.g.*, *id.*; Matthew Tokson, *Artificial Intelligence and the Anti-Authoritarian Fourth Amendment*, 27 U. PA. J. CONST. L. 1067 (2025); Paul Mozur et al., *'They Are Watching': Inside Russia's Vast Surveillance State,* N.Y. TIMES (Sept. 22 2022) https://www.nytimes.com/interactive/2022/09/22/technology/russia-putin-surveillance-spying.html.

39 MANTISELLI, *supra* note 37 at 1.

40 *Id.*; Sheera Frenkel & Aaron Krolik, *How ICE Already Knows Who Minneapolis Protesters Are*, N.Y. TIMES (Jan. 30, 2026), https://www.nytimes.com/2026/01/30/technology/tech-ice-facial-recognition-palantir.html; Alex Marthews & Catherine Tucker, *The Impact of Online*

undermining authoritarian regimes via the internet, authoritarian regimes have used the internet to interfere with democratic processes in other countries.[41] Further, the decentralized nature of internet communication has arguably degraded the global information environment in ways that are likely to undermine democracy.[42] Not every optimistic prediction about the internet's social impact was wrong. But the early optimism about the internet's pro-democracy effects looks quaint in retrospect. Predicting the social effects of transformative new technologies is exceptionally difficult, and optimistic predictions, no less than skeptical ones, may prove false over time.

II. HISTORICAL SKEPTICISM ABOUT RISK

Many potential catastrophic risks never come to fruition, like the supposed risk of excessive global population growth that arose briefly in the late 1960s.[43] Many hurricane and tornado warnings turn out to be false alarms, leading residents of disaster zones associated with prior false alarms to take future warnings less seriously.[44] Likewise, most potential pandemics that garner media attention wind up affecting relatively few people.[45] But, obviously, major pandemics and catastrophes sometimes do occur. Moreover, failing to heed warnings because of prior false alarms seems to

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Surveillance on Behavior, in THE CAMBRIDGE HANDBOOK OF SURVEILLANCE LAW 437 (David Gray & Stephen E. Henderson eds., 2017); Jonathan W. Penney, *Chilling Effects: Online Surveillance and Wikipedia Use*, 31 BERKELEY TECH L.J. 117, 125–29 (2016).

41 S. REP. NO. 116-290 (2020).

42 Bolane Olaniran & Indi Williams, *Social Media Effects: Hijacking Democracy and Civility in Civic Engagement, in* PLATFORMS, PROTESTS, AND THE CHALLENGE OF NETWORKED DEMOCRACY 77, 83 (John M. Jones & Mark Trice eds., 2020).

43 *See, e.g.*, PAUL R. ELRICH, THE POPULATION BOMB (1970); Thomas C. Peterson et. al., *The Myth of the 1970s Global Cooling Scientific Consensus*, 89 BULL. AM. METEOROLOGICAL SOC'Y 1325, 1325–26, 1333 (2008).

44 Kevin M. Simmons & Daniel Sutter, *False Alarms, Tornado Warnings, and Tornado Casualties*, 1 WEATHER CLIMATE & SOC'Y 38, 39-40

45 *See, e.g.,* Bethany Saxon et al., *Ebola and the Rhetoric of US Newspapers: Assessing Quality Risk Communication in Public Health Emergencies*, 22 J. RISK RSCH. 1309, 1309-1310 (2019) (documenting that U.S. media characterized the outbreak as "apocalyptic" despite the threat being largely confined to West Africa and resulting in only four domestic diagnoses and one death); Celine Klemm et. al., *Swine flu and hype: a systematic review of media dramatization of the H1N1 influenza pandemic*, 19 J. RISK RSCH. 1, 1-2, 14-15 (2016) (finding that although H1N1 caused approximately 18,000 deaths, which is far fewer than the 250,000-500,000 seasonal deaths from influenza, media coverage generated disproportionate and widespread public fear).

substantially increase people's risk of death when disaster strikes.[46]

The upshot is that even massive, obvious-in-retrospect catastrophic risks are sometimes met with excessive skepticism. Famously, for decades, there was substantial uncertainty and skepticism surrounding human-caused global warming. Only as the reality of global warming gradually became obvious—long after substantial global warming had begun—did skepticism about its existence largely fade away. As of today, global warming has already caused millions of additional heat-related deaths worldwide; is projected to cause millions more deaths due to heat, disease, and flooding; and produces extensive environmental damage and habitat loss.[47]

The story of global warming begins with an energy-generating technology. Coal-burning energy generation has long played a foundational role in industrialization and modern economic development, and it remains deeply embedded in global infrastructure today.[48] Unfortunately, burning coal produces substantial quantities of carbon dioxide, along with many other pollutants.[49] Scientists have known that CO_2 traps heat and keeps it from escaping the Earth's atmosphere since the mid-1800s.[50] But until the 1950s, scientists believed that CO_2 could not actually warm Earth because the relevant parts of the atmosphere were already saturated with CO_2, so the addition of more carbon in the atmosphere would not trap any additional heat.[51] That scientific consensus turned out to be wrong. Gilbert Plass demonstrated in 1956 that adding CO2 to the upper layers of the atmosphere would trap additional heat.[52] He predicted that human-caused carbon

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46 *See* Simmons & Sutter, *supra* note 44, at 44. People exposed to prior false alarms tend to die from disasters at higher rates than those not exposed to false alarms. *Id.*

47 *See, e.g.*, A. M. Vicedo-Cabrera, et al., *The burden of heat-related mortality attributable to recent human-induced climate change*, 11 NATURE CLIMATE CHANGE 492 (2021); *Climate change*, WHO (Oct. 12, 2023), https://www.who.int/news-room/fact-sheets/detail/climate-change-and-health; *Climate change impacts*, NOAA, https://www.noaa.gov/education/resource-collections/climate/climate-change-impacts (last visited June 1, 2026).

48 Robert C. Allen, *Backward into the Future: The Shift to Coal and Implications for the Next Energy Transition*, 50 ENERGY POL'Y 17, 17–18 (2012); Mikalai Filonchyk et al., *Global Greenhouse Gas Emissions from Coal-Fired Power Plants*, 228 RES. CONSERVATION & RECYCLING 108808, at 1 (2026) ("Coal-fired power plants remain a key source of electricity globally, generating approximately 36 % of worldwide power output.").

49 *E.g., id.* ("Simultaneously, they represent the largest anthropogenic source of CO2 emissions, accounting for over 40 % of global fossil fuel combustion emissions.").

50 ERIC M. CONWAY & NAOMI ORESKES, MERCHANTS OF DOUBT 170 (2010).

51 SPENCER R. WEART, THE DISCOVERY OF GLOBAL WARMING 23 (2003).

52 Spencer Weart, *Global Warming: How Skepticism Became Denial*, 67 BULL. OF THE ATOMIC SCIENTISTS 41, 43 (2011); Weart, *supra* note 51, at 23.

emissions would substantially increase the global temperature.[53] Other scientists reinforced Plass's work over the next several years, establishing that the oceans would not absorb all of humanity's extra carbon,[54] and that CO2 levels were rising substantially.[55] Nonetheless, most scientists continued to ignore the potential for global warming, and many posited that mankind's emissions and actions were unlikely to ever disrupt the natural balance, given the Earth's powerful compensatory mechanisms.[56] Evidence of a warming trend, like a dramatic reduction in Arctic ice thickness, was largely dismissed.[57]

With the advent of computer-based climate models in the late 1970s, scientists began to observe that adding CO2 to the atmosphere increased modeled global temperature, and they predicted substantial real-world temperature increases.[58] But most scientists cautioned that that models were imperfect and that the existence of future global warming remained highly uncertain.[59] In the early 1980s, this skeptical consensus remained in place even as human-caused global warming began to impact global temperatures, beginning the steep upward slope in temperature that continues to the present day.[60] Rather, most experts emphasized the continued uncertainty of future temperatures and noted that the historical record showed substantial natural fluctuations.[61] The hot years of the 1980s might just be another natural blip, they concluded.[62]

In the late 1980s, scientists began to pay far more attention to climate change, and publications on the topic began to take off.[63] Political entities began to pay more attention as well, and in 1988 the United Nations created the Intergovernmental Panel on Climate Change (IPCC).[64] The IPCC's first report, issued in 1990, acknowledged that the globe had been warming,

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[53] G. N. Plass, *Carbon Dioxide and the Climate*, 44 AM. SCIENTIST 302, 302-316 (1956); Weart *supra* note 51, at 23.
[54] Weart, *supra* note 52, at 43; Weart, *supra* note 51, at 28.
[55] Weart, *supra* note 52, at 43; Weart, *supra* note 51, at 35–36.
[56] Weart, *supra* note 52, at 43, (citing BJ Mason, *Has the Weather Gone Mad?*, 177 THE NEW REPUBLIC, 21-23 (1977)); Weart, *supra* note 51, at 89, n.7.
[57] Weart, *supra* note 51, at 41.
[58] Weart, *supra* note 52, at 43.
[59] Weart, *supra* note 51, at 100, 115.
[60] Weart, *supra* note 51, at 118; P. D. Jones et al., *Global Temperature Variations Between 1861 and 1984*, 322 NATURE 430-434 (1986).
[61] Weart, *supra* note 51, at 118.
[62] *Id.*
[63] *Id.* at 152.
[64] *Id.*

possibly due to human causes, and predicted that human emissions would cause further global warming of several degrees Celsius by 2050.[65] Though the report noted the uncertainty of the future, it suggested several means through which government should begin to reduce climate change risk. Other prominent reports issued similar calls to action.[66] All were largely ignored by policymakers and the public, even as scientific views began to solidify.[67] In the early 1990s, surveys showed that a majority of climate experts predicted that significant global warming was likely, though with only a moderate degree of confidence.[68] International bodies began to reach early agreements to limit carbon emissions.[69] By the early 2000s, scientists became less ambivalent, and the IPCC's 2007 report concluded that the evidence of human-caused global warming was "unequivocal."[70]

Throughout this later period, several skeptical scientists questioned the existence of human-caused global warming, and these skeptics were amplified by fossil fuel interests seeking to avoid regulation.[71] The conservative George C. Marshall Institute issued an influential report in 1989 blaming recent warming trends on fluctuations in the Sun's output.[72] Bill Nierenberg, the retired director of the Scripps Institution on Oceanography, presented the report to executive officials, persuading the Bush administration to halt action on climate regulation.[73] Climate modeler Patrick J. Michaels, physicist Fred Seitz, retired government official Fred Singer, and others attacked the IPCC's 1995 finding that human activity had been demonstrated to be a cause of some global warming, launching both

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65 INTERGOVERNMENTAL PANEL ON CLIMATE CHANGE, CLIMATE CHANGE: THE IPCC 1990 AND 1992 ASSESSMENTS 74, 111 (1992).

66 *See A Brief History of Climate Change Discoveries*, UK RSCH. & INNOVATION, https://www.discover.ukri.org/a-brief-history-of-climate-change-discoveries/index.html (for a discussion on other IPCC reports). *See also 1990s Climate Change*, SOC. SCI. RSCH. COUNCIL, https://100.ssrc.org/1990s-climate-change (for a discussion on the Social Science Research Council research committee on Global Environmental Change).

67 Weart, *supra* note 51, at 157–58.

68 Weart, *supra* note 51, at 158–69.

69 *See, e.g.*, ORESKES & CONWAY, *supra* note 50, at 197; *United Nations Framework Convention on Climate Change*, UNITED NATIONS, 1992, http://unfccc.int/resource/docs/convkp/conveng.pdf).

70 ORESKES & CONWAY, *supra* note 50, at 169 (citing *Summary for Policy Makers in Climate Change 2008, the Physical Science Basis, Contribution of Working Group I to the Fourth Assessment Report of the Intergovernmental Panel on Climate Change*, INTERGOVERNMENTAL PANEL ON CLIMATE CHANGE, 2007, at 8, https://www.ipcc.ch/report/ar4/wg1).

71 Weart, *supra* note 51, at 160.

72 *See* ORESKES & CONWAY, *supra* note 50, at 186.

73 *Id.* (citing Leslie Roberts, *Global Warming: Blaming the Sun*, 246 SCI. 992-993 (1989).)

substantive and ad hominem attacks.[74] Richard Lindzen, an atmospheric physicist at MIT, argued that cloud thickness would likely increase due to warmer temperatures, blocking more sunlight and preventing further warming.[75] Although climate change skeptics were few in number and generally far less qualified to opine on the subject than proponents of the theory of human-caused global warming, mainstream media sources generally portrayed the opposing camps as similar and awarded their views roughly equal time.[76]

Global warming is a gradual process that gave scientists ample time to discover and confirm its existence. No firm consensus emerged until decades after the warming had begun. Even when most scientists finally reached agreement, a handful of skeptics influenced the public debate over the phenomenon, aided by industry actors seeking to avoid regulation and the media's tendency to report both sides of a controversy, no matter how lopsided.

Although AI's catastrophic risks are at present less certain than global warming and its catastrophic risks, several similarities with AI can be observed. As in the early 1990s with global warming, today, a majority of technological experts in the field agree that AI poses catastrophic risks to humanity, but acknowledge that their conclusions remain uncertain.[77] Numerous prominent experts and other scientists have raised concerns about AI's catastrophic risks,[78] but there are also prominent skeptics,[79] and

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[74] ORESKES & CONWAY, *supra* note 50, at 208 (citing Frederick Seitz, *A Major Deception on "Global Warming"*, WALL STREET J. 1996.)

[75] Weart, *supra* note 52, at 47; Weart, *supra* note 51, at 168–69.

[76] Weart, *supra* note 52, at 46; Maxwell T. Boykoff & Jules M. Boykoff, *Balance as Bias: Global Warming and the US Prestige Press*, 14 GLOBAL ENV. CHANGE 125-136 (2004); Aaron M. McCright & Riley E. Dunlap, *Challenging Global Warming as a Social Problem: An Analysis of the Conservative Movement's Counter-claim*, 47 SOC. PROBLEM 499-522 (2000); Weart, *supra* note 51, at 180 n.5.

[77] *See* Grace, et al., *supra* note 7, at 9:1.

[78] *See, e.g.*, Matt Egan, *AI Could Pose 'Extinction-Level' Threat to Humans and the US Must Intervene, State Dept.-Commissioner Report Warns*, CNN (Mar. 12, 2024), https://www.cnn.com/2024/03/12/business/artificial-intelligence-ai-report-extinction/index.html; Cade Metz, *'The Godfather of A.I.' Leaves Google and Warns of Danger Ahead*, N.Y. TIMES (May 4, 2023), https://www.nytimes.com/2023/05/01/technology/ai-google-chatbot-engineer-quitshinton.html; Yoshua Bengio et al., *Pause Giant AI Experiments: An Open Letter*, FUTURE LIFE INST. (Mar. 22, 2023), https://futureoflife.org/open-letter/pause-giant-aiexperiments.

[79] *See, e.g.*, Narayanan & Kapoor, *supra* note 3; Chris Williams, *AI Guru Ng: Fearing a Rise of Killer Robots Is Like Worrying About Overpopulation on Mars*, THE REGISTER (Mar. 19, 2015),

media depictions of the ongoing debate tend not to note the growing consensus of experts regarding the existence of significant catastrophic risks.[80] One substantial difference is that AI harms may occur far less gradually than global warming, which has been happening for nearly fifty years.[81] In any event, history suggests that decisionmakers may be prone to overlook even obvious global risks when some uncertainty exists, even in the face of emerging expert consensus. This may be especially likely when powerful and politically connected industries are motivated to promote skepticism and delay regulatory responses.

III. SUPERPOWER ARMS RACES

History also contains lessons about the risks of technological arms races between competing superpowers. Perhaps the most popular argument against meaningful AI regulation asserts that the United States is locked into an AI arms race with China, which the US must win at all costs.[82] According to this line of reasoning, the importance of the AI arms race justifies disregarding safety and deregulating AI companies almost entirely.[83] This argument often focuses on the potential military advantages of advanced AI, which, proponents claim, may confer a substantial strategic benefit on

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https://www.theregister.com/2015/03/19/andrew_ng_baidu_ai; Grace Dean, *One of the "Godfathers" of AI Says Concerns the Technology Could Pose a Threat to Humanity Are "Preposterously Ridiculous,"* BUS. INSIDER (June 15, 2023), https://www.businessinsider.com/yann-lecun-artificial-intelligence-generative-ai-threaten-humanity-existential-risk-2023-6.

[80] *See* Grace, et al., *supra* note 7, at 9:10.

[81] *See supra* note 60.

[82] *See, e.g.*, INTERESTING TIMES, *JD Vance on His Faith and Trump's Most Controversial Policies*, N.Y. TIMES (May 21, 2025), https://www.nytimes.com/2025/05/21/opinion/jd-vance-pope-trump-immigration.html; R. David Edelman et al., *How will AI influence US-China relations in the next 5 years?*, BROOKINGS (June 18, 2025), https://www.brookings.edu/articles/how-will-ai-influence-us-china-relations-in-the-next-5-years; *Winning the AI Race: Strengthening U.S. Capabilities in Computing and Innovation Before the S. Comm. on Com., Sci., and Transp.*, 119th Cong. (2025); Justin Sherman, *Reframing the U.S.-China AI "Arms Race"*, NEW AMERICA (Mar. 6, 2019), https://www.newamerica.org/cybersecurity-initiative/reports/essay-reframing-the-us-china-ai-arms-race/why-us-china-ai-competition-matters.

[83] *See, e.g.*, Memorandum from Secretary of War Pete Hegseth for Senior Pentagon Leadership Commanders of the Combatant Command Defense Agency and Dow Field Activity Directors 4 (Jan. 9, 2026) [hereinafter Hegseth Memorandum], https://media.defense.gov/2026/Jan/12/2003855671/-1/-1/0/artificial-intelligence-strategy-for-the-department-of-war.pdf.; David E. Sanger, *Vance Envisions America-First A.I. Race*, N.Y. TIMES Feb. 12, 2025, at B1.

whichever nation leads the way in AI development.[84]

Yet the most recent technological arms race between competing military superpowers suggests that gaining an advantage in a peacetime arms race confers almost no actual benefits on the "winner." During the Cold War, America led the nuclear race against the Soviet Union. But its lead was of little strategic value, and it quickly evaporated as the Soviets stole or reverse-engineered American designs and built their own nuclear weapons. The United States gained almost nothing from winning the nuclear race, while the race itself imposed massive downside risks on the world at large.[85] In the decades since the US and Soviet Union developed massive stockpiles of nuclear weapons, there have been several accidents that nearly lead to global thermonuclear war and the deaths of billions.[86] These catastrophes were avoided only through remarkable luck and the heroism of a few key individuals.[87]

A. The Thermonuclear Race

The nuclear arms race began in World War II, with the promise of decisive strategic advantage through superior weaponry.[88] The United States

[84] *See, e.g.*, *Remarks by Secretary Esper at National Security Commission on Artificial Intelligence Public Conference*, U.S. DEP'T OF WAR (Nov. 5, 2019), https://www.war.gov/News/Transcripts/Transcript/Article/2011960/remarks-by-secretary-esper-at-national-security-commission-on-artificial-intell; *With China's DeepSeek, US tech fears red threat*, THE CHRONICLE (Jan. 30, 2025), https://www.thechronicle.com.au/news/breaking-news/with-chinas-deepseek-us-tech-fears-red-threat/news-story/3c0bffe6ee0f6f54ea85c52bafbd6523; *infra* note 139 and accompanying text.

[85] *See* RICHARD RHODES, DARK SUN: THE MAKING OF THE HYDROGEN BOMB 505, 523 (1995). Although the United States initially enjoyed a monopoly on nuclear weapons, that advantage quickly eroded as the Soviet Union developed atomic and then thermonuclear weapons of its own. The primary effect of these breakthroughs was not enduring strategic advantage but an ever-growing increase in national defense expenditures and, most critically, the number of civilians at risk of dying in a nuclear exchange. *See also* Hans Kristensen et al., Status Of World Nuclear Forces, FED'N AM. SCIENTISTS (Mar. 26, 2025), https://fas.org/initiative/status-world-nuclear-forces (reporting that By the mid-global nuclear stockpiles peaked at roughly 70,000 nuclear warheads, enough to cause a global nuclear winter lasting decades).

[86] TOBY ORD, THE PRECIPICE 96–99 (2020).

[87] *See, e.g.*, *id.* at 4–5, 96–97.

[88] *See* Tokson & Arbel, *Artificial Intelligence and Existential Risk*, 59 CONN. L. REV. (forthcoming); JOHN LEWIS GADDIS, THE UNITED STATES AND THE ORIGINS OF THE COLD WAR, 1941-1947, 244–46 (1972).

tested its first nuclear bomb in 1945, and deployed nuclear bombs shortly thereafter in Hiroshima and Nagasaki.[89] But the United States' monopoly on nuclear weapons evaporated relatively quickly. The Soviet Union tested its first atomic bomb in 1949, using a design closely modeled on the American bomb, developed in part through intelligence gathered by spies inside the Manhattan Project.[90]

The Soviet nuclear detonation caught American leaders by surprise and, combined with the Chinese Communist Revolution of 1949, stoked fears in many Americans of rising global communist power. A debate arose among nuclear scientists and in the Truman White House as to whether the U.S. should pursue thermonuclear weapons.[91] Thermonuclear weapons (commonly known as H-bombs) are, at present, the most destructive weapons ever created.[92] Unlike the first generation of atomic bombs, H-bombs create thermonuclear fusion, causing a larger and more destructive nuclear reaction.[93] Following the USSR's successful nuclear test, advocates of developing a thermonuclear weapon argued that a new superweapon could allow the US to preserve its lead in a nuclear arms race against the Soviets.[94]

The Atomic Energy Commission's General Advisory Committee, chaired by Robert Oppenheimer, issued a report in October of 1949 recommending that the United States decline to pursue thermonuclear weapons.[95] The debate raged on inside the Truman Administration, and President Truman turned to a special subcommittee of the National Security Council (NSC) for advice.[96] The subcommittee was divided, with some members urging the pursuit of an international agreement on nuclear energy with the Soviet Union, and others urging rapid development of thermonuclear weapons.[97] The pro-development faction ultimately prevailed. In early 1950, President Truman directed the Armed Forces to develop thermonuclear weapons, and construction soon after began on the nuclear reactors necessary to produce fuel for the new weapons.[98]

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[89] RHODES, *supra* note 85, at 175–76.

[90] STEVEN J. ZALOGA, THE KREMLIN'S NUCLEAR SWORD 5–11 (2002).

[91] ATOMIC HERITAGE FOUNDATION, *Hydrogen Bomb – 1950* (June 19. 2014), https://ahf.nuclearmuseum.org/ahf/history/hydrogen-bomb-1950.

[92] *Id.*

[93] RHODES, *supra* note 85, at 247–248.

[94] *Id.* at 381.

[95] HERBERT YORK, THE ADVISORS: OPPENHEIMER, TELLER, AND THE SUPERBOMB 48–52 (1976).

[96] RHODES, *supra* note 85, at 404.

[97] *Id.* at 404–05.

[98] *Id.* at 407.

In early 1951, the United States conducted a series of tests to verify that their leading design for a fusion bomb was viable. Physicists Edward Teller and Stanislaw Ulam developed an effective two-stage design using a small fission reaction to ignite a larger fusion reaction.[99] The first thermonuclear bomb was tested in November 1952 in the Marshall Islands. It produced the energy equivalent of 10 megatons of TNT, roughly 1000 times as much as the bomb dropped on Hiroshima.[100]

The United States had won the race to develop thermonuclear weapons. The politicians and physicists who had argued that the US need to produce a much bigger bomb than the atomic bomb possessed by the Soviets had achieved their goal.[101] And yet, the result was not an enduring geopolitical advantage, or even meaningful short-term leverage against the Soviet Union. The result was that the Soviets succesfully tested their first fusion bomb in August of 1953, less than a year after the first US test.

The Soviet Union had several spies working within the Manhattan Project.[102] From the beginning of the Project, Soviet intelligence had received regular updates on America's atomic weapons development programs, including the beginnings of the thermonuclear weapons program.[103] Nuclear physicist and Soviet spy Klaus Fuchs reported to Soviet intelligence about a secret conference on thermonuclear bomb development at Los Alamos National Laboratory in 1946.[104] Inspired and aided by this report, the Soviets launched their own efforts to develop a thermonuclear weapon in order to counter the coming American weapon.[105]

At the Physics Institute of the Soviet Academy of Sciences, the USSR's leading theoretical physicists began designing fusion weapons.[106] In 1948, Pioneering Soviet scientist Andrei Sakharov developed a concept for a relatively simple fusion bomb that layered various light and heavy elements around a plutonium core.[107] The Soviets successfully tested a fusion bomb based on this design in August 1953, less than a year after the United States' first test (and delayed substantially by Joseph Stalin's desire to build two

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[99] *Id.* at 468, 472.
[100] ATOMIC HERITAGE FOUNDATION, *supra* note 91.
[101] *Id.*
[102] RHODES, *supra* note 85, at 123–125.
[103] ATOMIC HERITAGE FOUNDATION, *Soviet Hydrogen Bomb Program* (Aug. 8, 2014), https://ahf.nuclearmuseum.org/ahf/history/soviet-hydrogen-bomb-program/
[104] RHODES, *supra* note 85, at 254-55
[105] ATOMIC HERITAGE FOUNDATION, *supra* note 103; RHODES, *supra* note 85, at 254–57, 332–33.
[106] *Id.* at 332.
[107] *Id.* at 334.

bombs before testing, one to test and one to counter potential American aggression).[108] In 1955, using a design likely copied from the American H-bomb, the Soviets successfully tested a larger thermonuclear bomb.[109] Ironically, early Soviet research may have helped hasten the development of the United States's thermonuclear bomb, as intelligence reports on Soviet research informed scientists in the United States about various ideas and breakthroughs, much as the Soviets had been informed about US developments.[110]

In the following years, both the United States and the Soviet Union produced large numbers of nuclear weapons as well as long-range ballistic missiles and various air-based and sea-based delivery vehicles, growing their nuclear stockpiles in an ongoing competition with no discernable end. By the mid-1960s, both countries had enough nuclear weapons to completely obliterate each other, as well as the ability to launch a devastating attack even after sustaining a full assualt from the other nation.[111] In October of 1962, the Cuban Missile Crisis would nearly cause a global thermonuclear war; it was the first and most famous of several near-global-catastrophes of the thermonuclear era.[112] Eventually, the futility of the nuclear arms race and the reduction in direct geopolitical conflict between the US and USSR resulted in a period of relative détente, and the two sides reached several treaties that allowed for a limit to and then reduction in the number of nuclear weapons.[113] The Cold War ended when the Soviet Union collapsed politically and economically, a collapse hastened somewhat by military spending demands but largely caused by other factors.[114] Russia and the United States still maintain massive stockpiles of nuclear weapons, and while the risk of global thermonuclear war has substantially decreased, these stockpiles still create substantial risks of catastrophic military escalation or catastrophic accidents.[115]

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[108] *Id.* at 524.
[109] *Id.* at 569.
[110] *Id.* at 465.
[111] *Id.* at 575–76.
[112] RHODES, *supra* note 85, at 572–73
[113] ORD, *supra* note 86, at 100–01.
[114] *See, e.g.*, David L. Hoffmann, *The Soviet Collapse*, ORIGINS: CURRENT EVENTS IN HISTORICAL PERSPECTIVE (Dec. 2021), https://origins.osu.edu/article/soviet-collapse-yeltsin-putin-gorbachev-russia; Celeste A. Wallander, *Western Policy and the Demise of the Soviet Union*, 5 J. COLD WAR STUD. 137 (2003).
[115] ORD, *supra* note 86, at 27, 101–02.

B. The Perils of Racing

An H-bomb's destructive radius varies based on the size of the bomb and typically covers a mile or two, but larger bombs can cover far larger areas. The largest H-bomb ever tested, the Soviet Union's Tsar Bomba, imposed severe damage in a 150-mile radius from the blast site.[116] In 1986, at the peak of nuclear weapon stockpiling, the United States possessed roughly 23,300 nuclear warheads and the Soviet Union possessed roughly 40,100.[117] The nuclear fallout of even a small exchange of modern nuclear weapons would put millions of lives at risk by generating enormous amounts of atmospheric soot, leading to widespread famines.[118] A full-scale nuclear war between superpowers would likely result in billions of deaths and could possibly result in total human extinction.[119]

Despite the fact that neither the Soviet Union nor the United States ever intentionally launched a nuclear attack on the other superpower, global thermonuclear war nearly occurred many times. The most famous of these, indeed the only one with which most people are likely to be familiar, was the Cuban Missile Crisis. In July of 1962, Soviet leader Nikita Khrushchev agreed with Fidel Castro to place Soviet nuclear missiles in Cuba to deter a future US invasion.[120] US spy planes spotted the missile launch facilities in Cuba in October 1962.[121] Many of President Kennedy's advisors suggested that he conduct an air strike on Cuba followed by a ground invasion, a course of action that likely would have resulted in global nuclear war. Kennedy instead reached a secret agreement with Khrushchev, agreeing that the US would dismantle its nuclear weapons in Turkey and Italy and the Soviets would

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116 NATIONAL WWII MUSEUM, *Tsar Bomba: The Largest Atomic Test in World History* (Aug. 29, 2020), https://www.nationalww2museum.org/war/articles/tsar-bomba-largest-atomic-test-world-history; U.S. DEP'T OF HEALTH & HUM. SERVS. *Damage Zones after a Nuclear Detonation: Idealized Maps* (Mar. 23, 2026) https://remm.hhs.gov/zones_nucleardetonation.htm#:~:text=Damage%20is%20caused%20by%20shocks,a%2010%20KT%20nuclear%20explosion.

117 Kristensen et al., *supra* note 85 (reporting that Russia has roughly 5,400 nuclear weapons and the US has roughly 5,000 nuclear weapons).

118 Jonas Jägermeyr, et al., *A regional nuclear conflict would compromise global food security*, 117 PNAS 7071 (Mar. 16, 2020).

119 ORD, *supra* note 86, at 99.

120 *The Cuban Missile Crisis, October 1962*, OFFICE OF THE HISTORIAN, U.S. DEP'T OF STATE, https://history.state.gov/milestones/1961-1968/cuban-missile-crisis (last visited June 2, 2026).

121 *Id.*

dismantle their nuclear weapons in Cuba.[122] Kennedy's willingness to ignore his advisors and his preference for peaceful solutions narrowly averted nuclear war.

Perhaps the most perilous nuclear miss occurred in 1983, when U.S. and U.S.S.R. nuclear stockpiles were near their combined peak.[123] Tensions between the countries were especially high, and Soviet leaders were convinced that the United States was planning a preemptive nuclear attack.[124] On September 26, 1983, Lieutenant Colonel Stanislav Petrov was on duty in a command center outside of Moscow where the Soviet military monitored its early-warning satellites. The missile warning siren began to shriek and the large computer screen glowed red, showing the Russian word for "launch."[125] The radar showed that five intercontinental ballistic missles had launched from an American base.[126] The missiles would hit Soviet targets in a matter of minutes.[127] The Soviet Union's strategic posture dictated that Soviet missiles be launched in retaliation, and global thermonuclear war begin.[128]

Only heroic human intervention saved the world from catastrophe. Rather than report the launch to his commanders, likely triggering massive Soviet retaliation and American re-retaliation, Colonel Petrov reported a system malfunction, guessing correctly that the alarm was a satellite error rather than a real nuclear launch.[129] A less savvy or more paranoid officer, or

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[122] *Id.*

[123] See Kristensen et al., *supra* note 85.

[124] *E.g.*, John T. Correll, *The Euromissle Showdown*, AIR & SPACE FORCES MAG. (Feb. 1, 2020), https://www.airandspaceforces.com/article/the-euromissile-showdown; Sewell Chan, *Stanislav Petrov, Soviet Officer Who Helped Avert Nuclear War, Is Dead at* 77, N.Y. TIMES (Sept. 18, 2017), https://www.nytimes.com/2017/09/18/world/europe/stanislav-petrov-nuclear-war-dead.html (noting that tensions were high following the USSR's deployment of nuclear missiles capable of striking any point in Western Europe and NATO's counter-deployment of nuclear missiles in capable of striking many points in the Soviet Union); Marc Bennetts, *Soviet Officer Who Averted Cold War Nuclear Disaster Dies Aged 77*, GUARDIAN (Sept. 18, 2017), https://www.theguardian.com/world/2017/sep/18/soviet-officer-who-averted-cold-war-nuclear-disaster-dies-aged-77.

[125] *Id.*

[126] *Id.*; Chan, *supra* note 124.

[127] Bennetts, *supra* note 124.

[128] *See* Rose Gottemoeller, *The Standstill Conundrum: The Advent of Second-Strike Vulnerability and Options to Address It*, 4 TEX. NAT'L SEC. REV. 115 (2021).

[129] *See*, *e.g.,* Bennetts, *supra* note 124. Petrov's guess was in part based on the low number of missiles detected and in part based on the lack of confirmation of launches by Soviet radar systems, although the latter could only detect missiles until they were substantially closer to the U.S.S.R. *Id.*; David Hoffman, *'I Had A Funny Feeling in My Gut'*, Wash. Post (Feb. 10, 1999), https://www.washingtonpost.com/wp-srv/inatl/longterm/coldwar/shatter021099b.htm.

a different guess in a situation that Petrov himself called "50-50," would likely have caused global nuclear war, mass death on an unprecedented scale, and the possibility of total human extinction.[130]

Several other nuclear near misses occurred during and after the Cold War, some involving false alarms and computer errors[131] and others involving confused and overly aggressive military officers.[132] As a result of an arms race between superpowers, the world was repeatedly brought to the brink of catastrophe, saved only by a handful of human decisionmakers who ignored the advice of military hawks and the erroneous warnings of their computer systems.

The primary lesson of the last superpower arms race is that the strategic benefits of winning such a race are likely to be both slight and temporary. Designing and manufacturing a powerful weapon makes it more likely that an opposing superpower will learn how to make such a weapon, and far more likely that they will attempt to do so. Artificial intelligence is a digital technology, and American breakthroughs in AI are especially likely to be distributed, copied, or reverse-engineered by China and other nations.[133] The idea that a United States company or military division will develop, test, and deploy an AI technology that substantially bolsters US military capability and that the US will somehow be able to retain this technological advantage indefinitely so as to realize an enduring (or even short-term) geopolitical advantage, is likely unsound. Rather, if the US is able to win the race to develop a groundbreaking new AI-military technology, China is likely to catch up shortly after, with both sides building up stockpiles of arms while

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[130] *See* Chan, *supra* note 124. *See also* ORD, *supra* note 86, 98–100 (discussing the prospect of total extinction resulting from nuclear war).

[131] For example, in 1979, US nuclear command centers prepared to launch a nuclear attack in response to their computers showing a full scale Soviet strike heading towards the US. The Soviet strike turned out to be a training scenario that was mistakenly sent out to the actual nuclear computer system. ORD, *supra* note 86, at 96. Even after the Cold War, in 1995, a Russian radar system mistook a Norwegian scientific rocket studying the northern lights for a US nuclear missile, nearly triggering nuclear retaliation. ORD, *supra* note 86, at 96-97.

[132] In 1962, a Soviet submarine captain gave the order to fire a nuclear missile at US forces after his sub was attacked by a US warship. A second officer agreed, but the third officer whose agreement was required refused to grant permission, and the submarine instead surrendered. ORD, *supra* note 86, at 4–5.

[133] *See, e.g.*, *Protecting Our Edge: Trade Secrets and the Global AI Arms Race Before the Subcomm. on Cts., Intell. Prop., and the Internet of the H. Comm. on the Judiciary*, 119th Cong. (2025) (statement by Benjamin Jensen, Dir. and Senior Fellow, CSIS Futures Lab); Billy Perrigo, *Every AI Datacenter Is Vulnerable to Chinese Espionage, Report Says*, TIME (Apr. 22, 2025), https://time.com/7279123/ai-datacenter-superintelligence-china-trump-report;

the dangers of a catastrophic accident or destructive conflict escalate accordingly.[134]

The implications of the harrowing history of nuclear near-misses are simple, but profound. Creating highly destructive weapons can give rise to significant risks of global catastrophe. Americans have largely forgotten the existential anxiety of the Cold War and are largely ignorant of the more worrisome near-misses of nuclear history. We are at risk of dramatically underestimating the harms of building and stockpiling advanced weapons in a competition between superpowers. Unimaginable global catastrophes were averted during the last superpower arms race only through dumb luck,[135] the heroism of individual humans,[136] and the fact that humans had authority to

[134] *See* Tokson & Arbel, *supra* note 88. To be sure, the benefits of winning an arms race may differ sharply in wartime compared to peacetime, especially in a major war involving territorial invasion. The Manhattan Project was launched in the midst of an all-encompassing world war, and racing to develop a nuclear bomb before the Nazis did was a perfectly reasonable course of action. This holds true even though the Nazis never invested in anything like a Manhattan Project of their own nor came anywhere close to developing an actual atomic weapon, and despite the fact that they surrendered before the US developed such a weapon. If the United States were to become embroiled in a major shooting war with China, then racing to develop effective new weapons of whatever kind would likely be justified. But central to that justification is the prospect of a finish line: using those new weapons against enemy targets in order to win, and end, the war. In peacetime, however, there is no war to end, and no finish line to reach—or at least none that is morally or strategically acceptable. The only sure way to "win" a peacetime arms race in any lasting way would be to use one's superweapon against the opposing country in order to destroy it or credibly threaten to use the weapon in order to conquer it. Preemptively unleashing a superweapon against a peaceful geopolitical competitor in order to conquer or destroy it would be a moral abomination and a geopolitical disaster. *See* Tokson & Arbel *supra* note 88. It would also be likely to provoke devastating retaliation, because both Chinese and US nuclear systems are neither networked nor easily disrupted by cyber attacks, and they may prove impossible for even highly advanced AI systems to neutralize. *See* Simon Goldstein & Peter N. Salib, *Nuclear Deterrence in the Age of AGI*, LAWFARE (Mar. 26, 2025), https://www.lawfaremedia.org/article/nuclear-deterrence-in-the-age-of-agi. Looking to history, preemptive attack on a peacetime rival using highly destructive weapons was briefly and hypothetically contemplated in the wake of America's development of the atomic bomb, but it was wisely rejected out of hand by President Truman, no pacifist when it came to the use of nuclear weapons. *See* RHODES, *supra* note 85, at 225–26.

[135] *See* ORD, *supra* note 86, at 4–5. A nuclear submarine captain and a Soviet political officer concurred on launching a nuclear missile at US forces during the US blockade of Cuba. By sheer luck, their submarine was the one out of four submarines in their flotilla that carried the commander of the flotilla, and he did not concur, averting the nuclear strike and, most likely, full-scale nuclear war. *Id.*

[136] *See* ORD, *supra* note 86, at 96-97. During the September 1983 "Autumn Equinox

disregard computer warnings that normally would have compelled them to initiate a catastrophic conflict.[137] Should advanced AI weapons be built in order to gain a perceived advantage in a superpower competition with China, all three of these factors might be absent. The first two largely occur via random chance, and as to the third, competitive pressures are likely to push military commanders to grant AI systems greater and greater autonomy over launch commands and other military decisions.[138] The current Administration has expressly ordered the integration of AI into a variety of military systems and emphasized that safety and perfect control are low priorities in the context of a military "race" in which "speed wins."[139] The interaction of AI and existing nuclear infrastructure may be especially dangerous, and such integration is already occurring with little public scrutiny.[140] The logic of an AI military race, like that of a nuclear race, is likely to lend itself to rapid escalation, retaliatory attack dynamics, and substantial risks of catastrophic accidents.[141]

CONCLUSION

As this Essay demonstrates, history does not support complacency about the future impacts of AI. Throughout history, optimists have often been wrong about the social ramifications of new technologies or the strategic benefits of building new weapons. Skeptics have often

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Incident," Soviet early-warning systems falsely indicated that several U.S. intercontinental ballistic missiles had been launched. Stanislav Petrov, the duty officer responsible for reporting the warning, reported that it was a false alarm despite some uncertainty. *See* RHODES, *supra* note 85, at 574-576. During the Cuban Missile Crisis, President Kennedy rejected recommendations from Strategic Air Command officials to launch military strikes against Cuba and instead pursued a negotiated resolution with the Soviet Union. *See supra* notes 120–122 and accompanying text.

[137] ORD, *supra* note 86, at 96–97.

[138] Francis Fukuyama, *Delegation and Destruction*, PERSUASION: FRANKLY FUKUYAMA (June 13, 2025), https://www.persuasion.community/p/delegation-and-destruction.

[139] *See* Hegseth Memorandum, *supra* note 83. Relatedly, Secretary Hegseth contends that "the risks of not moving fast enough outweigh the risks of imperfect alignment [between humans and AI.]" *Id. See also* David E. Sanger, *Vance Envisions America-First A.I. Race*, N.Y. TIMES Feb. 12, 2025, at B1 (quoting Vice President J.D. Vance stating that "[t]he A.I. future is not going to be won by hand-wringing about safety").

[140] *See, e.g.*, Greg Hadley, *AI 'Will Enhance' Nuclear Command and Control, Says STRATCOM Boss*, AIR & SPACE FORCES MAG. (Oct. 28, 2024), https://www.airandspaceforces.com/stratcom-boss-ai-nuclear-command-control.

[141] *See* Tokson & Arbel, *supra* note 88, at 19–21.

underestimated the likelihood of novel innovations and their impacts on humanity. Many transformative technologies were underestimated or misunderstood in their early stages, and AI is likely no different. The wisest course in forecasting AI's development is to acknowledge that the future of novel technologies is often impossible to predict before it unfolds. The challenge for policymakers is to address the potential harms of AI despite the inevitable uncertainty that surrounds its future development.[142]

New technologies don't always underperform, aren't always beneficial, and sometimes threaten disaster. In the absence of a historical basis for complacency, the best present-day policymakers can do is to survey a wide variety of experts and thinkers, acting on the basis of informed estimates and diverse sources of insight. A single authority will not suffice, no matter how brilliant; even Albert Einstein got the future wrong.[143]

Rejecting history-based complacency may help policymakers to see the unique risks and benefits of AI more clearly. Indeed, AI differs from many prior technologies in that a majority of experts in the field already agree that it poses substantial, possibly catastrophic risks to humanity.[144] Several of the inventors of modern AI, along with numerous prominent scientists from a variety of fields, have sounded the alarm about these risks and many other, more immediate harms.[145] The fact that new technologies have generally been beneficial provides little comfort in the AI context. The lessons of technological history suggest that new technologies can exceed conventional expectations, have harmful effects on society, cause serious environmental harms, and even risk unimaginable global catastrophe. Learning from this

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142 *See* Yonathan Arbel, Matthew Tokson & Albert Lin, *Systemic Regulation of Artificial Intelligence*, 56 ARIZ. ST. L.J. 545, 584-586 (2024) (arguing that AI regulation must address novel and difficult-to-predict harms arising from future advances in AI capabilities).

143 *See supra* note 14.

144 *See* Grace, et al., *supra* note 7, at 9:10.

145 *See, e.g.*, Matt Egan, *AI Could Pose 'Extinction-Level' Threat to Humans and the US Must Intervene, State Dept.-Commissioner Report Warns*, CNN (Mar. 12, 2024), https://www.cnn.com/2024/03/12/business/artificial-intelligence-ai-report-extinction/index.html; Cade Metz, *'The Godfather of A.I.' Leaves Google and Warns of Danger Ahead*, N.Y. TIMES (May 4, 2023), https://www.nytimes.com/2023/05/01/technology/ai-google-chatbot-engineer-quits-hinton.html; Hideyuki Matsumi & Daniel J. Solove, *The Prediction Society: Algorithms and the Problems of Forecasting the Future*, 2025 U. ILL. L. REV. 1, 22-25 (2023) (algorithmic discrimination); Cassandra Cross, *Using Artificial Intelligence (AI) and Deepfakes to Deceive Victims: The Need to Rethink Current Romance Fraud Prevention Messaging,* 24 CRIME PREVENTION & CMTY. SAFETY 30, 35-36 (2022) (AI-enabled fraud through deepfakes and synthetic identities); Cameron F. Kerry, *Protecting Privacy in an AI-Driven World*, BROOKINGS (Feb. 10, 2020), https://www .brookings.edu/articles/protecting-privacy-in-an-ai-driven-world (privacy harms).

history could save us a world of trouble in the future.